9)

III. MODEL OF A COIN - 30 MARKS

You have a coin. However, you don't know its model. While the coin is two faced like any other coin, the probability P with which tossing it gives heads is unknown. You know that P must take a value in the set $\{0.1, 0.3, 0.5, 0.9\}$. You start with the belief that P takes the values with equal probability (the prior PMF of P). Let heads be valued 100 and tails be valued 0. What is the expected value obtained from tossing the coin?

You perform an experiment in which you toss the coin five times and get three heads and two tails.

1) How does the outcome of the experiment change your belief (updated PMF) about P?
2) Let heads be valued 100 and tails be valued 0. What is the expected value obtained from tossing the coin?

$$P[P=p \mid A] = \frac{P[P=p, A]}{P[A]}$$

[Your updated belief]

$$= \frac{P[A \mid P=p]\, P[P=p]}{P[A]}$$

$$= \frac{P[A \mid P=p]\, P[P=p]}{\sum_{i=1}^{4} P[A \mid P=p_i]\, P[P=p_i]}$$

$$= \frac{p^3(1-p)^2\, P[P=p]}{\sum_{i=1}^{4} p_i^3(1-p_i)^2\, P[P=p_i]}$$

$$P[P=0.3 \mid A] = \frac{(0.3)^3(0.7)^2\,(1/4)}{\frac{1}{4}\left[(0.1)^3(0.9)^2 + (0.3)^3(0.7)^2 + (0.5)^3(0.5)^2 + (0.9)^3(0.1)^2\right]}$$

and so on.

1

On Calculating Expected Values:

Consider when your belief is as given by the prior PMF, that is the PMF before the experiment was performed.

The PMF is

$$P[P=x] = \begin{cases} 0.25 & x = 0.1, 0.3, 0.5, 0.9 \\ 0 & \text{otherwise.} \end{cases}$$

Define X to be the RV that determines the outcome of a coin toss.

$$S_X = \{0, 100\}$$

$$E[X] = 0\, P[X=0] + 100\, P[X=100]$$
$$= 100\, P[X=100]$$

$$P[X=100] = P[X=100, P=0.1] + P[X=100, P=0.3] + P[X=100, P=0.5] + P[X=100, P=0.9]$$

$$= P[X=100 \mid P=0.1]\, P[P=0.1] + P[X=100 \mid P=0.3]\, P[P=0.3] + P[X=100 \mid P=0.5]\, P[P=0.5] + P[X=100 \mid P=0.9]\, P[P=0.9]$$

If you are told that the coin gives heads with probability 0.1, that is you are told that $P=0.1$, then

$$P[X=100 \mid P=0.1] = 0.1.$$

$$\therefore P[X=100] = (0.1)\tfrac{1}{4} + (0.3)\tfrac{1}{4} + (0.5)\tfrac{1}{4} + (0.9)\tfrac{1}{4} \;//$$

$$E[X] = 100\, P[X=100]$$ [See above]

What about the expected value after you have gotten the results of the 5 coin tosses?

Note that the range space S_P does not change. Range space never changes. However, the probabilities you associate with the outcomes 0.1, 0.3, 0.5, 0.9 in the range space are updated on seeing the outcome of five coin tosses. An example of how the probability is updated is given by equation 1 above.

So now you have a "new" PMF.

Again, consider X.

$$S_X = \{0, 100\}$$

$$E[X] = 100\, P[X=100]$$

where

$$P[X=100] = \sum_{p_i \in S_P} P[X=100, P=p_i]$$

$$= \sum_{p_i \in S_P} P[X=100 \mid P=p_i]\, \underbrace{P[P=p_i]}$$

This here is your "new" PMF!